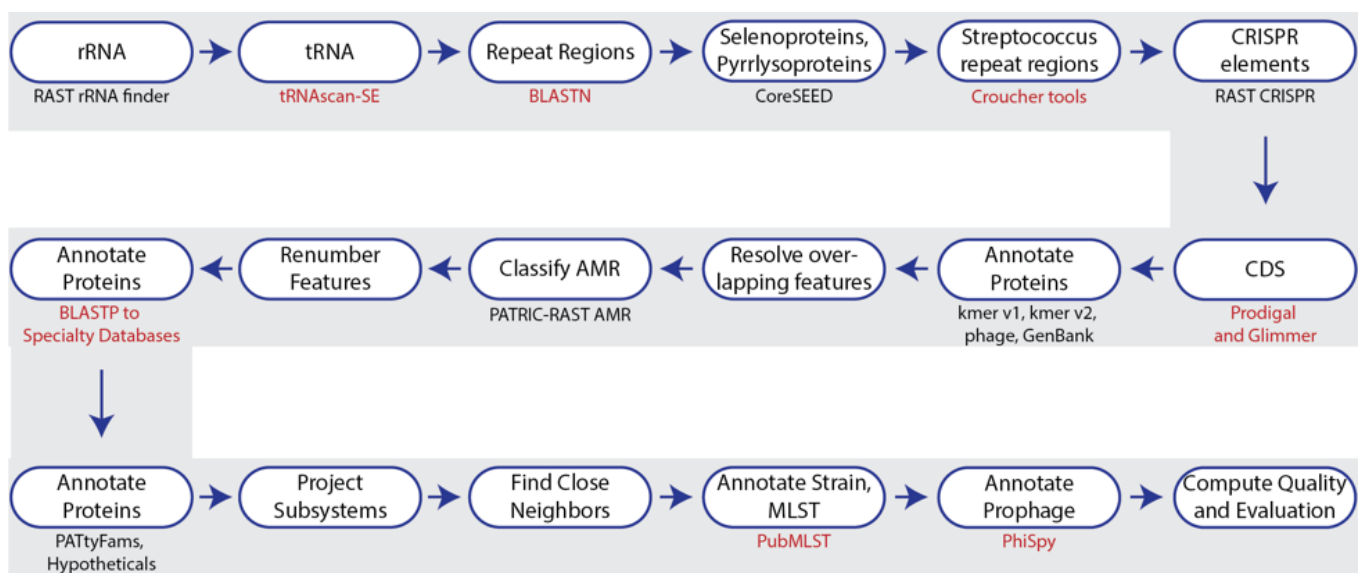


Genome Annotation Service

Revised 09 December 2024

Genome annotation is the process of identifying functional elements along the sequence of a genome. The Genome Annotation Service in BV-BRC uses the RAST tool kit (RASTtk)[1] to annotate genomic features in bacteria and archaeal genomes, the Viral Genome ORF Reader (VIGOR4)[2,3] or Mat_peptide[4] to annotate viral genomes, and PHANOTATE[5,6] to annotate bacteriophage genomes. All public genomes in BV-BRC have been consistently annotated with these services. Researchers can also submit their own private genome to the annotation service, where it will be deposited into their private workspace for their perusal and comparison with other BV-BRC genomes using BV-BRC analysis tools and services, e.g., Phylogenetic Tree, Genome Alignment, Comparative Systems, Similar Genome Finder and others.

RASTtk Annotation of Bacteria and Archaea

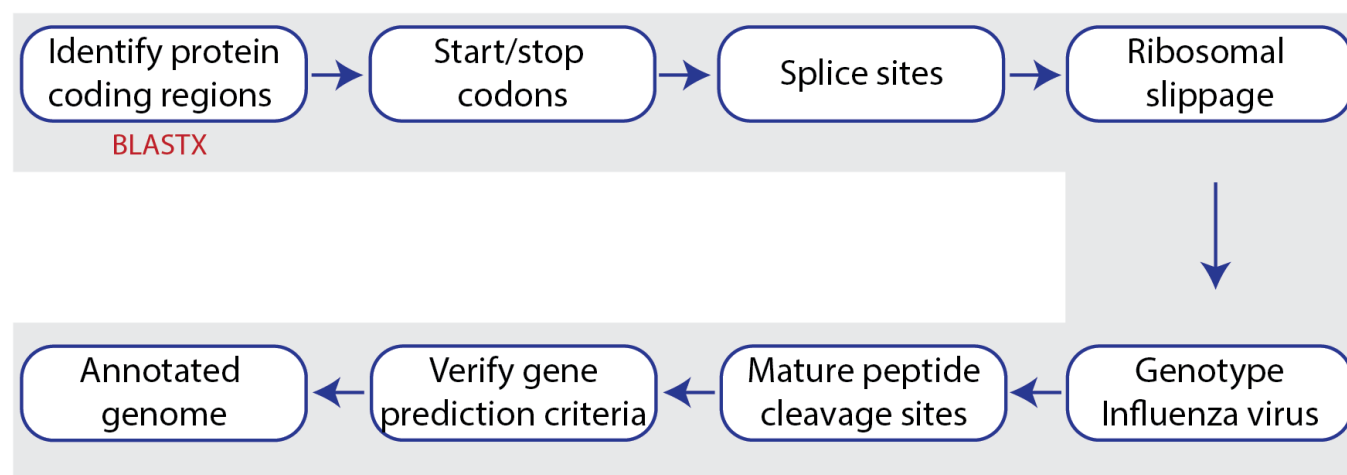


The bacterial and archaeal annotation service available in BV-BRC uses a modular, updated version of RAST[7] called the RAST toolkit (RASTtk)[1]. It includes algorithms that were developed by the RAST team and some that were developed by others and incorporated into the overall pipeline (seen in red in the figure above). tRNAscan-SE[8] is used to call the tRNA genes. BLASTN[9] is used to identify repeat regions within the genome, and tools by Croucher[10] are used to identify *Streptococcus* repeat regions. After repeat regions are identified, Prodigal[11], followed by Glimmer[12], are used to call coding sequences (CDS). Antimicrobial resistance is projected for a select group of genera based on a Adaboost machine learning[13], followed by an initial protein annotation event that involves taking every protein called in a genome and using BLAT[14] and BLASTP[15] to identify CDSs that have homology to proteins in specialty databases. Possible virulence factors are identified by blasting against a database containing proteins collected from the

Virulence Factor Database[15], Violins[17], and a special curation effort by the BV-BRC team[18]. Genes with homology to those identified as being involved in antimicrobial resistance are BLATed against proteins from the Comprehensive Antibiotic Resistance Database (CARD)[19], the National Database of Antibiotic Resistant Organisms (NDARO <https://www.ncbi.nlm.nih.gov/pathogens/antimicrobial-resistance/>) and a special curation of relevant proteins by BV-BRC curators[20]. Genes with homology to transporters are identified by searching against proteins from the Transporter Classification Database (TCDB)[21], and those similar to genes that have been identified as potential drug targets by comparison to proteins from DrugBank[22] and the Therapeutic Target Database (TTD)[23]. Protein families[24] are assigned, and then hypothetical proteins are identified. All proteins are then mapped to subsystems[25,26] when possible. PubMLST[27] is used to assign sequence types, and then PhiSpy[28] is used to find prophages in bacterial genomes.

The source code for RASTtk is available on Github (<https://github.com/SEEDtk/RASTtk>).

VIGOR4 Annotation of Viruses



The Viral Genome ORF Reader, known as VIGOR4[2,3], is a Java application to predict protein sequences encoded in viral genomes. The protein coding sequences are determined by sequence similarity searching against curated viral protein databases. VIGOR4 uses the VIGOR_DB project which currently has databases for the following viruses:

- Influenza (A & B for human, avian, and swine, and C for human)
- SARS-CoV-2, Sarbecovirus, Nobecovirus, Hibecovirus, Embecovirus, Merbecovirus, Wuhivirus
- West Nile Virus Lineage I/II, Zika Virus, Dengue Virus
- Chikungunya Virus
- Alphaviruses: Eastern Equine Encephalitis Virus, Venezuelan Equine Encephalitis Virus
- Respiratory Syncytial Virus
- Rotavirus A/B/C/F/G
- Enterovirus, Entovirus, Goukovirus, Horwuvirus, Hudivirus, Hudovirus, Ixovirus, Laulavirus, Lentinivirus, Mobuvirus, Phasivirus, Pidchovirus, Rubodvirus, Tenuivirus, Uukuvirus, Wenrivirus
- Lassa Mammarenavirus, Antennavirus, Bandavirus, Reptarenavirus

- Hantaviridae, Nairoviridae, Peribunyaviridae, Phlebovirus
- Sapovirus
- Monkeypox Virus
- Beidivirus
- Feravirus, Hartmanivirus, Inshuvirus, Jonvirus, Sawastrivirus
- Coguvirus, Cicadellivirus, Orthophasmavirus

The source code for VIGOR4 is available on GitHub (<https://github.com/JCVenterInstitute/VIGOR4>).

Mat_peptide Annotation of Viruses

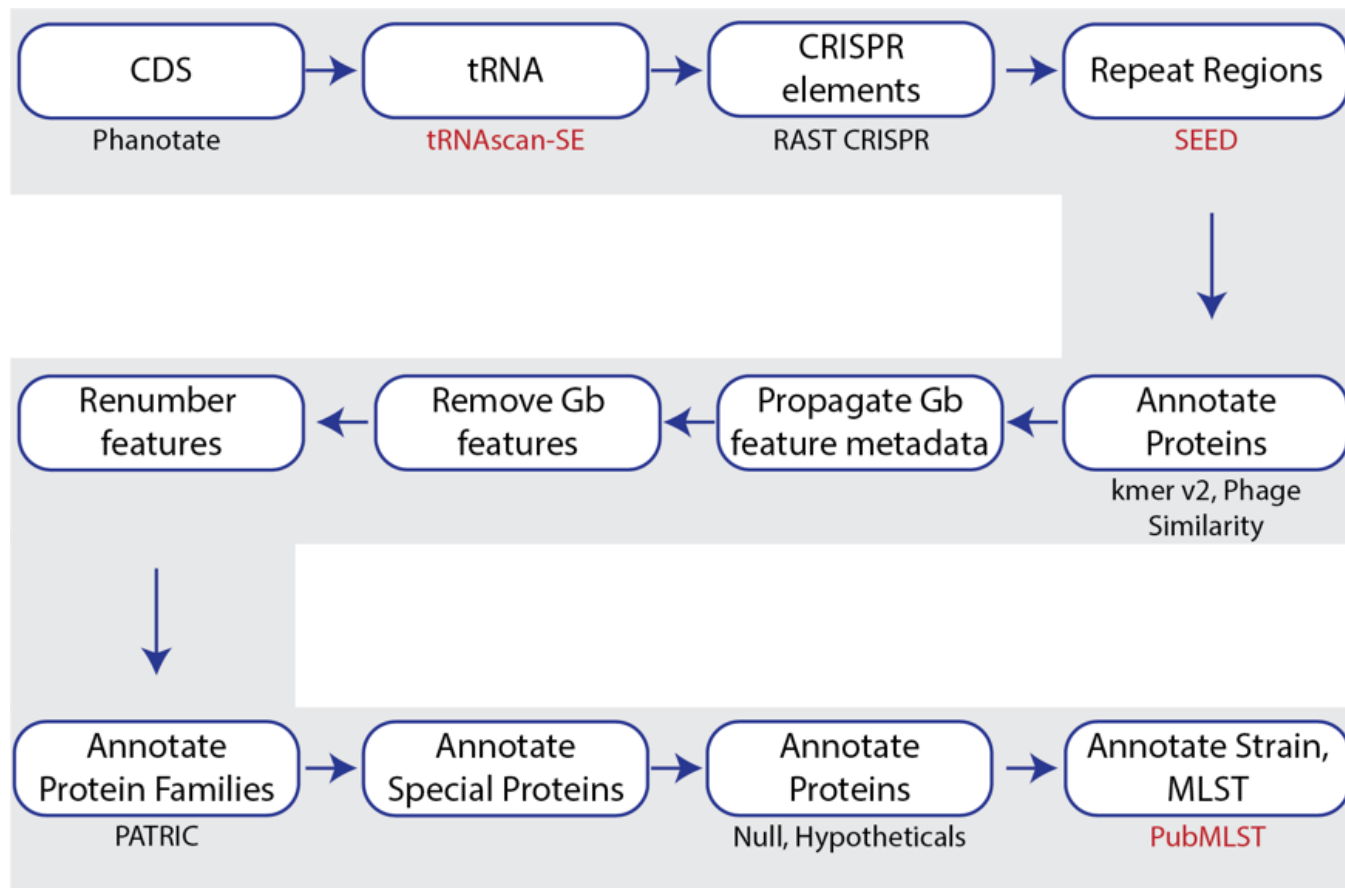


The Mat_peptide[4] pipeline is used to annotate 165 different species that are part of sixteen viral families:

- Arenaviridae
- Caliciviridae
- Flaviviridae
- Hantaviridae
- Nairoviridae
- Orthomyxoviridae
- Peribunyaviridae
- Phasmaviridae
- Phenuiviridae
- Picornaviridae
- Pneumoviridae
- Poxviridae
- Sedoreoviridae
- Togaviridae

Mat_peptides uses pairwise alignment with reference sequences. Mat_peptide calculates and transfers curated mature protein annotation positions from a reference to a target sequence. This pipeline is deployed when an appropriate viral taxon that cannot be annotated by Vigor4 is submitted to the BV-BRC Annotation service. The source code for Mat_peptide is available on GitHub (software at https://github.com/VirusBRC/vipr_mat_peptide).

PHANOTATE Annotation of Bacteriophages



PHANOTATE is a bacteriophage genome annotation pipeline [5,6] that identifies open-reading frames and then performs sequence similarity searches to annotate proteins. The source code for PHANOTATE is available on GitHub (<https://github.com/deprekate/PHANOTATE>).

Locating the Annotation Service App

1. At the top of any BV-BRC page, find the **Services** tab.



2. In the drop-down box, underneath **Genomics**, click on **Genome Annotation**.

The screenshot shows the BV-BRC website interface. The top navigation bar includes links for ORGANISMS, SEARCHES, TOOLS & SERVICES, WORKSPACES, HELP, and ABOUT. The main content area is titled 'BACTERIAL AND VIRAL BIOINFORMATICS' and includes a welcome message, a search bar, and a 'BROWSE' section with buttons for BACTERIA and VIRUS. A 'SEARCH' section is also present. The 'TOOLS & SERVICES' dropdown menu is open, showing various tools categorized into Genomics, Metagenomics, Transcriptomics, Utilities, Viral Tools, Protein Tools, and Outbreak Tracker. The 'Genome Annotation' option under the 'Genomics' category is highlighted with a red dashed box and a red arrow.

3. This will open up the Annotation Service landing page.

The screenshot shows the 'Genome Annotation' service landing page. The page title is 'Genome Annotation' with an information icon. The description states: 'The Genome Annotation Service provides annotation of genomic features using the RAST tool kit (RASTtk) for bacteria and VIGOR4 for viruses. The service accepts a FASTA formatted contig file and an annotation recipe based on taxonomy to provide an annotated genome. For further explanation, please see the Genome Annotation Service [Quick Reference Guide](#) and [Tutorial](#).' Below the description is a 'Parameters' form with the following fields:

- CONTIGS**: A file upload field with a dropdown menu and a folder icon.
- ANNOTATION RECIPE**: A dropdown menu with the option '--- Select Recipe ---'.
- TAXONOMY NAME**: A text input field with a question mark icon, containing the example 'e.g. Bacillus cereus'.
- TAXONOMY ID**: A dropdown menu.
- MY LABEL**: A text input field containing the example 'My identifier123'.
- OUTPUT NAME**: A text input field containing the example 'Taxonomy + My Label'.
- OUTPUT FOLDER**: A file upload field with a dropdown menu and a folder icon.

At the bottom of the form are two buttons: 'Reset' and 'Annotate'.

Selecting a contig file for annotation

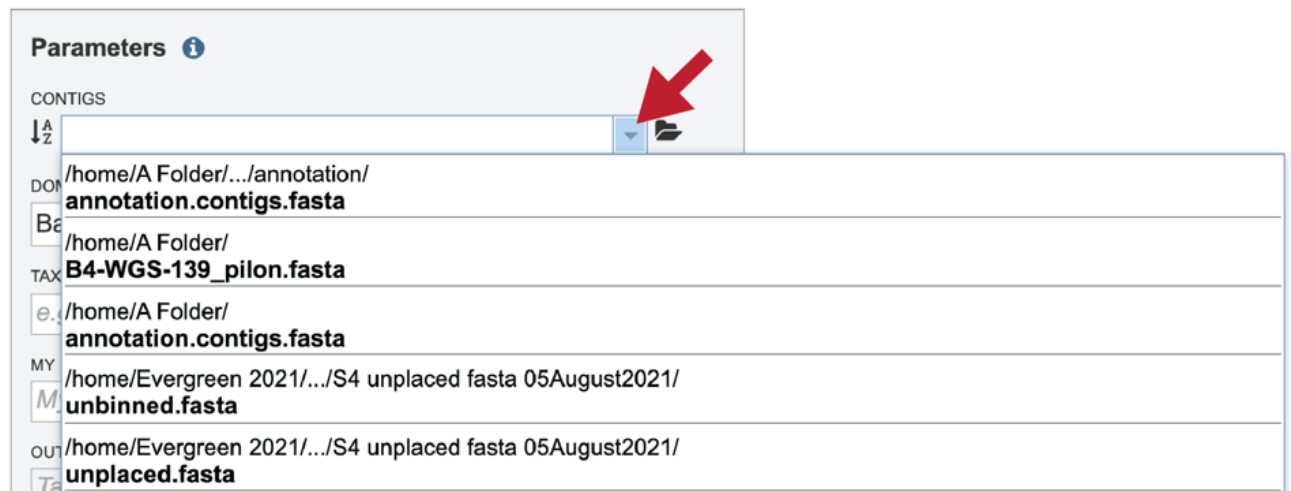
The starting point for any annotation is an assembly, which produces contigs. A contig (from the word "contiguous") is a series of overlapping DNA sequences used to make a physical map that reconstructs the original DNA sequence of a chromosome or a region of a chromosome. It is a stretch of DNA sequence encoded as A, G, C, T or N, typically ending in fasta or fa. The first line of a contig file begins with ">", followed by information on the contig. The second and subsequent line(s) contain the sequences.

```
>NODE_1_length_315326_cov_18.8348
```

```
GGCATCCAACGCCATTCATGGCCATATCAATGATTTTCTGGTGCGTACCGGGTTGAGAAG
CGGTGTAAGTGAAGTGCAGTTGCCATGTTTTACGGCAGTGAGAGCAGAGATAGCGCTGAT
GTCCGGCAGTGCTTTTGCCGTTACGCACCACCCCGTCAGTAGCTGAACAGGAGGGACAGC
TGATAGAAACAGAAGCCACTGGAGAACCTCAAAAACACCATCATACTAAATCAGTAAG
TTGGCAGCATCACCCGTATTGTCCATGCGATGAATAAAGGAAAAGTTATGAAACACAAGC
```

Contigs must be submitted to the annotation service. **Submitting a read file will not work.** Researchers can upload a contig file that was generated in BV-BRC, or one that they have assembled independently.

1. If an assembly was generated in BV-BRC the contigs can be selected by clicking on the down arrow at the end of the text box. This will show the contig files that are available.



Parameters ⓘ

CONTIGS

↓ A Z

DO /home/A Folder/.../annotation/
annotation.contigs.fasta

Be /home/A Folder/
B4-WGS-139_pilon.fasta

TAX /home/A Folder/
annotation.contigs.fasta

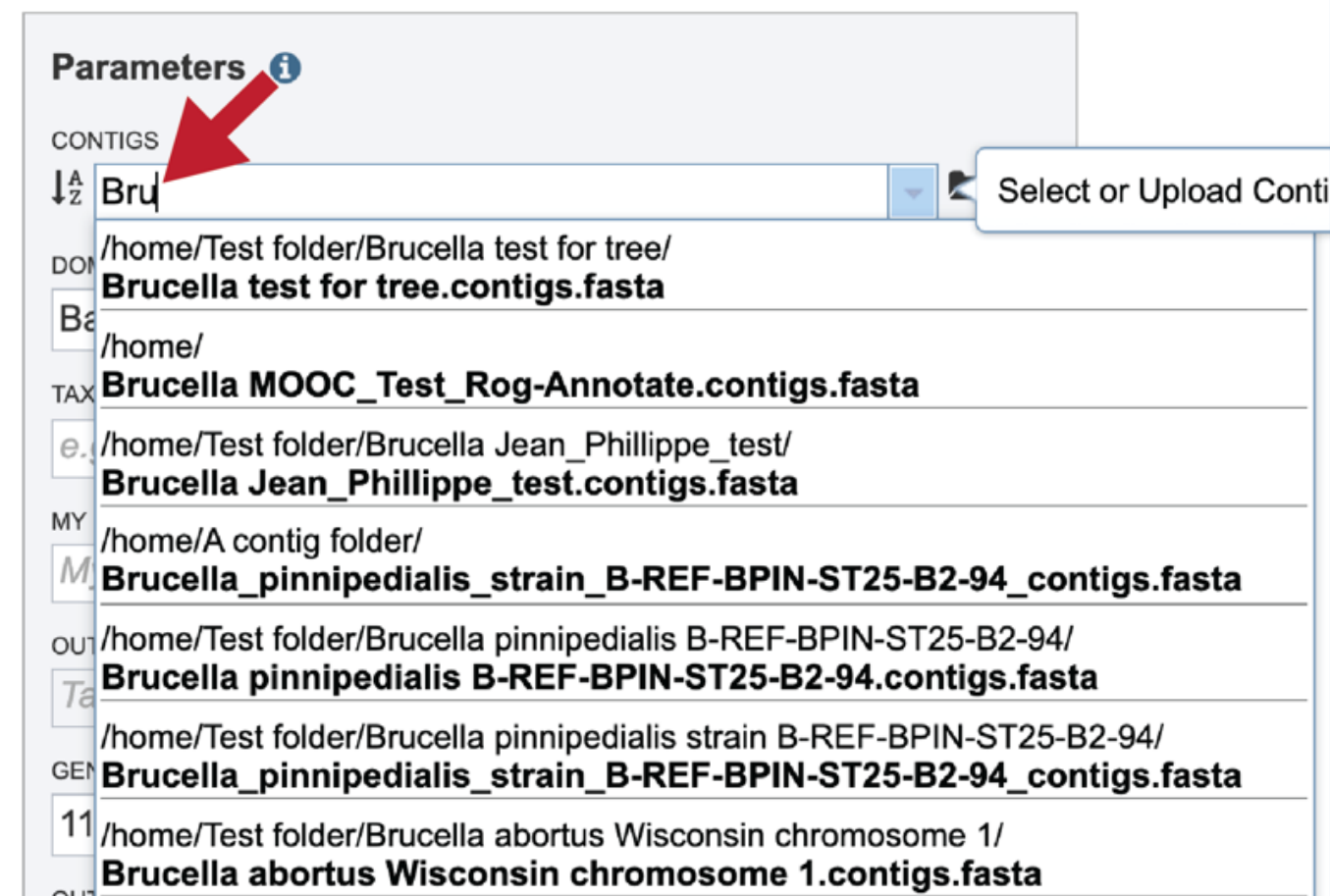
e. /home/Evergreen 2021/.../S4 unplaced fasta 05August2021/
unbinned.fasta

MY /home/Evergreen 2021/.../S4 unplaced fasta 05August2021/
unplaced.fasta

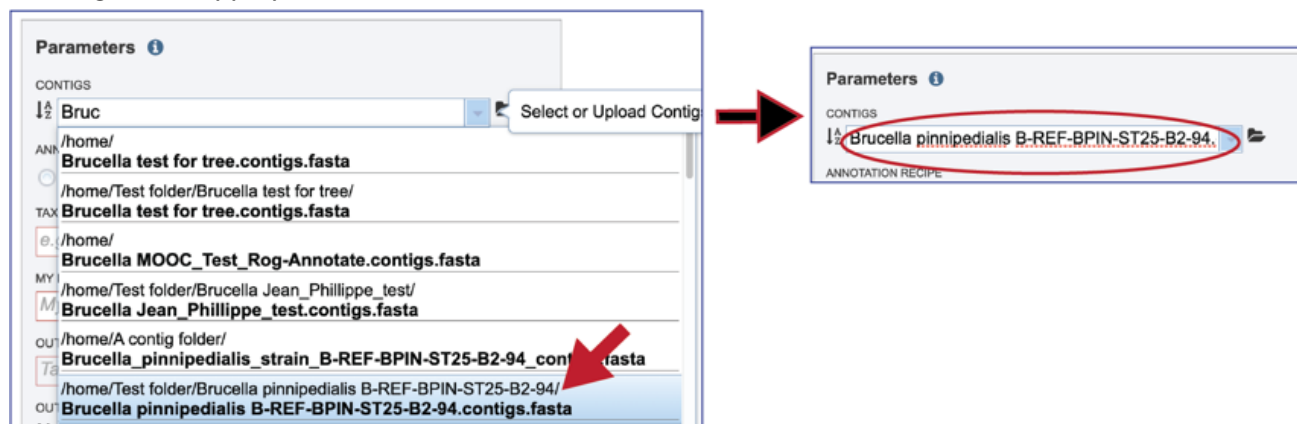
OUT /home/Evergreen 2021/.../S4 unplaced fasta 05August2021/
unplaced.fasta

Ta

2. If the name is not easily seen, begin typing the name in the text box. The search function will start looking for names that match the text that is entered.



3. Clicking on the appropriate name will fill the text box.



4. If an assembly has been generated outside of BV-BRC, click on the **Folder** at the end of the text box. If you want to upload data directly to your home directory, click on the icon with the arrow pointing up. This opens up a pop-up window where the files for upload can be selected. Click on the icon with the **arrow** pointing up.

Parameters ⓘ

CONTIGS

1/2

Choose or Upload a Workspace Object

Workspaces ⓘ

/ARWattam@patricbrc.org/home

Selection: None

Name	Size	Owner	Members	
Parent folder				
MRSP8468v4.fasta	2.6 MB	me	Only me	11/15/19, 11:40 AM
.SRR8773941 human gut metagenome Kraken		me	Only me	3/26/19, 11:07 AM
01-S1.fasta	5.4 MB	me	Only me	10/26/20, 1:46 PM
01_Yellow_0.fa	97.3 MB	me	Only me	11/4/20, 4:26 PM
02_Yellow_1.fa	50.8 MB	me	Only me	11/4/20, 4:34 PM
03_Green_0.fa	31.7 MB	me	Only me	11/4/20, 4:35 PM
04_June2020 Demo		me	Only me	6/4/20, 3:15 PM
04_Green_1.fa	49.0 MB	me	Only me	11/4/20, 4:36 PM
05_Blue_0.fa	46.6 MB	me	Only me	11/4/20, 4:37 PM
06_Blue_1.fa	174.8 MB	me	Only me	11/4/20, 4:38 PM
07_Nycodens_PMA.fa	89.6 MB	me	Only me	11/4/20, 4:39 PM
08_klondike_v1.fa	73.0 MB	me	Only me	11/4/20, 4:41 PM

☒ Show all files and folders ⓘ

Cancel OK

5. This opens a new window where the file you want to upload can be selected. Click on the **Select File** in the blue bar.

Upload files to Workspace

Browse Workspace

Upload file to: /ARWattam@patricbrc.org/home

Upload type: Contigs

Contigs must be provided in fasta format (typically .fa, .fasta, .fna). Genbank formatted files are not currently accepted.

☒ Restrict file selection to the common extensions for this file type:
.fa, .fasta, .faa, .fna

Select File

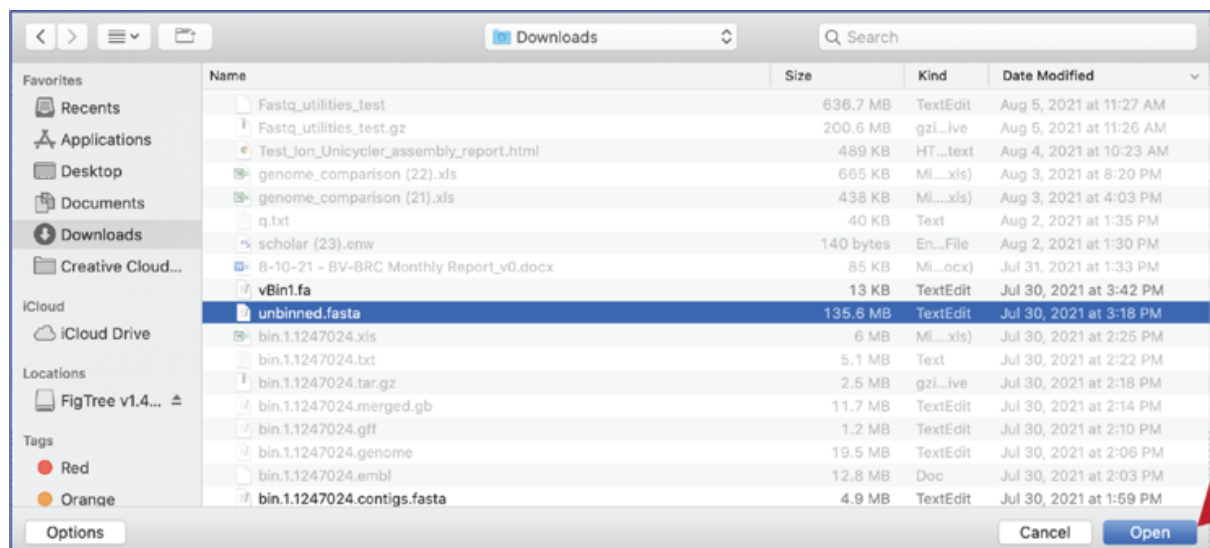
or

Drop files here.

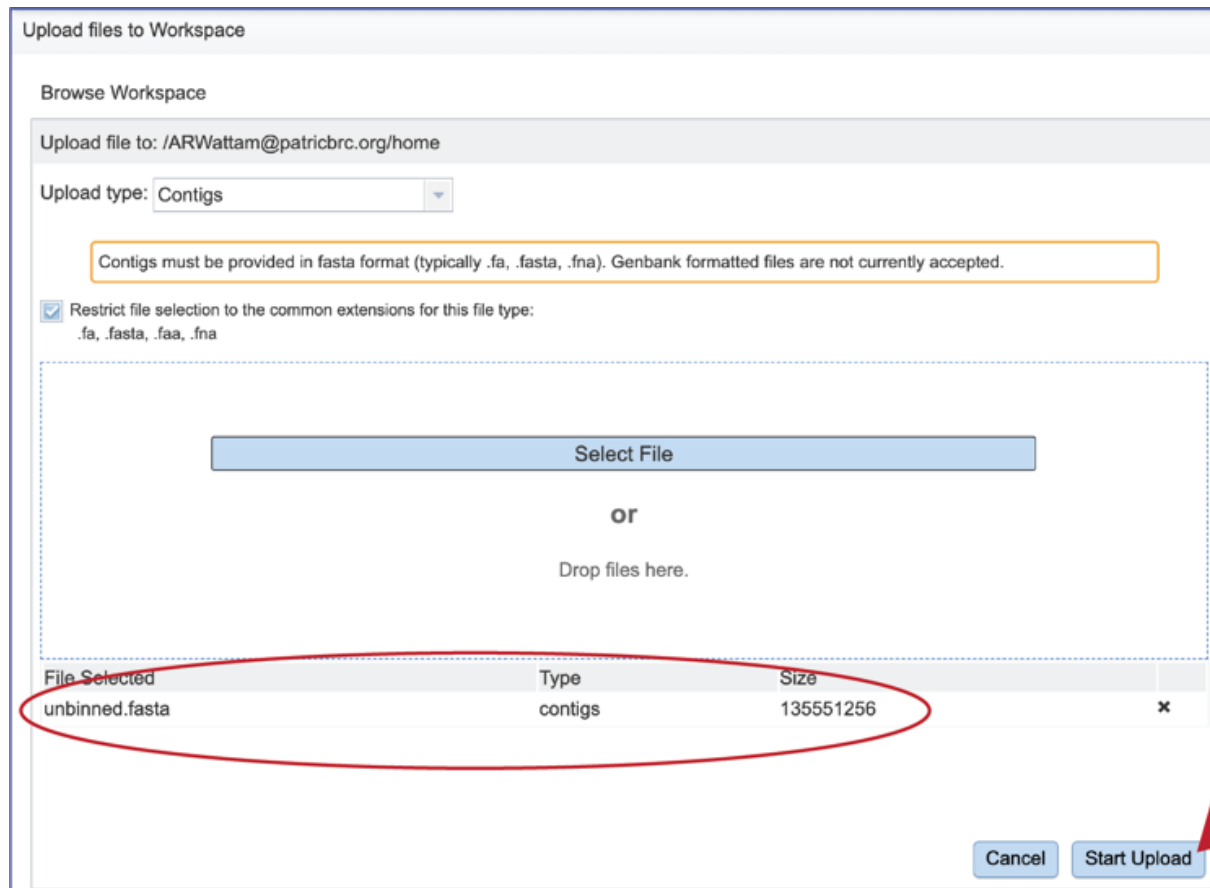
File Selected	Type	Size
None		

Cancel Start Upload

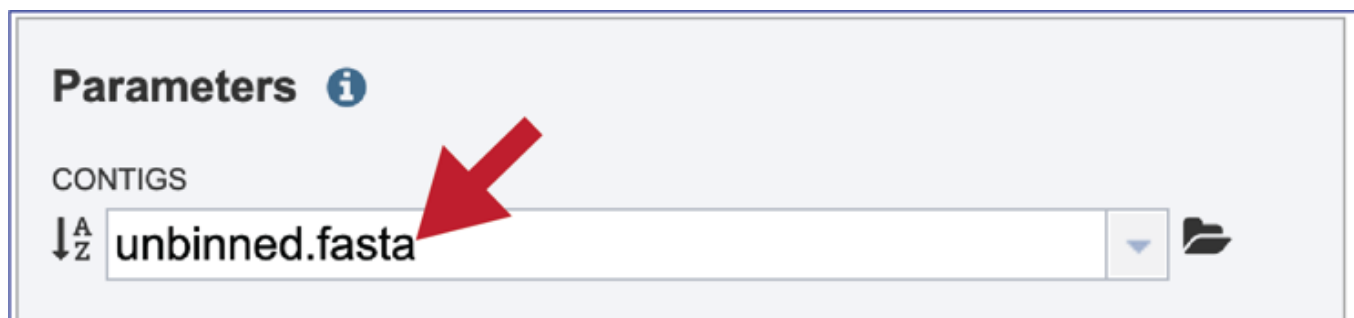
6. This will open a window that allows you to choose files that are stored on your computer. Select the file where you stored the contig file on your computer and click **Open**.



7. Once selected, it will autofill the name of the file. Click on the **Start Upload** button.

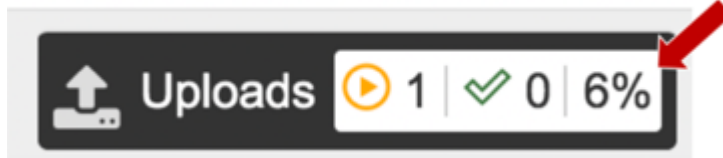


8. This will autofill the name of the document into the text box. This should be a quick process as contig files are generally small.

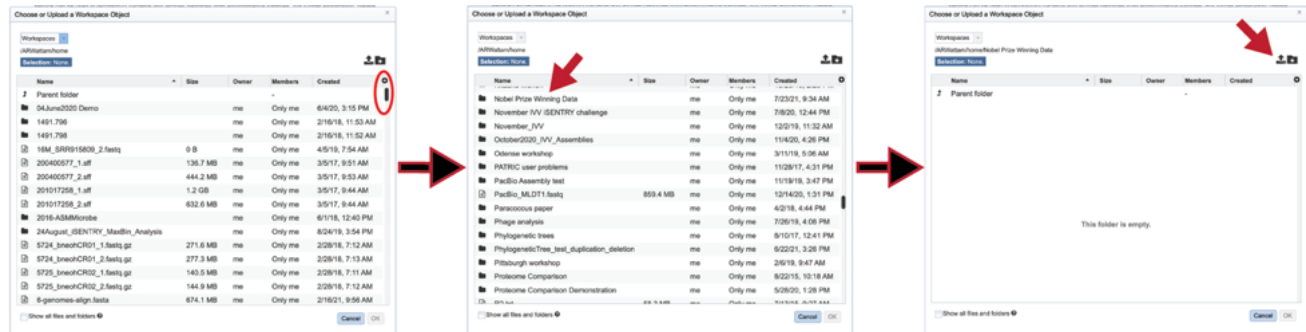


9. Progress of the upload can be seen in the **Uploads** information bar at the bottom of every BV-BRC page.

The three columns show completed jobs (first column), jobs in progress (second), and the percent completion of the jobs in progress (third). Wait until the upload is complete prior to submitting the job.



10. You can also navigate to a desired folder. Use the scroll bar at the left of the pop-up window to see all the data in your directory. When the desired folder is found, click on it. This will open that directory in the window, where you can use the upload icon to upload data directly to the selected folder.



Annotation Parameters

1. Annotation parameters must be selected next. BV-BRC provides annotation for Bacteria, Archaea and Bacteriophages. Bacteria and Archaea are annotated using the RASTtk pipeline. Bacteriophage genomes are annotated using the PHANOTATE pipeline. Viruses are annotated with VIGOR4 or Mat_peptide. To select a particular annotation strategy from one of those taxa, click on the check box preceding the correct strategy.

Parameters ?

CONTIGS

↓^A ↑^Z 📁

ANNOTATION RECIPE

☒ BACTERIA/ARCHAEA
 ☐ VIRUSES
 ☐ BACTERIOPHAGES

2. The taxonomic name must next be selected. Begin typing in the lowest ranked taxonomic name known for the sequenced isolate. For bacteria/archaea, try to get to Genus, if possible, so that the annotation will contain two types of protein families; global (cross-genus) and local (within genus)[24]. Once typing begins, a drop-down box will start showing taxonomic names that match the text entered.

[genus] *Bruchola*

3. Click on the most appropriate name, which will autofill the text box and also the corresponding Taxonomy ID. If the Taxonomy ID is known, that can be filled in first and the matching taxonomic name will be autofilled. BV-BRC provides two different version of protein families, which are called PATtyFams [24]. If a taxonomic level above Genus is selected, the annotation will only have global protein families (PGFams) assigned. If a genus or species is selected, the annotation will include both PGFams, and the local protein families (PLFams), which are genus specific.

234

4. Give the genome a unique name by entering text in the box underneath **My Label**. The name that is entered will appear in the **Output Name** in the lowest text box.

Parameters

CONTIGS

  **Brucella pinnipedialis B-REF-BPIN-ST25-B2-94.**

ANNOTATION RECIPE

☐ BACTERIA/ARCHAEA ☐ VIRUSES ☐ BACTERIOPHAGES

TAXONOMY NAME  TAXONOMY ID

Brucella **234**

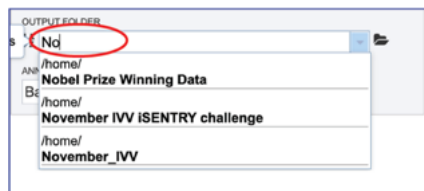
MY LAB

test

OUTPUT NAME

Brucella test

5. An output folder must be selected for the assembly job. Typing the name of the folder in the text box underneath the words **Output Folder** will show a drop-down box that shows close hits to the name. Clicking on the arrow at the end of the box will open a drop-down box that shows the most recently created folders. To find a previously created folder that is not seen in the drop-down box, click on the folder icon at the end of the text box. This will open a pop-up window that shows all the previously created folder, then follow the instructions at the top of this tutorial.





6. Once the input data and the parameters have been selected, the **Submit** button at the bottom of the page will turn blue. The annotation job will be submitted once this button is clicked. Once submitted, the job will enter the queue. You can check the status of your job by clicking on the **Jobs** monitor at the lower right.

Parameters 

CONTIGS

ANNOTATION RECIPE

☒ BACTERIA/ARCHAEA ☐ VIRUSES ☐ BACTERIOPHAGES

TAXONOMY NAME 



TAXONOMY ID



MY LABEL

OUTPUT NAME

OUTPUT FOLDER



Finding the completed Annotation job

1. There are two places to access a completed job in BV-BRC. Clicking on the **Jobs** icon at the bottom right of any page will open the list of jobs that have been submitted.

BV-BRC BETA ORGANISMS DATA SEARCHES SERVICES WORKSPACES HELP

NEW: Check out our [SARS-CoV-2 Variants and Lineages of Concern](#) and [Genome Assembly and Annotation Service](#).

BACTERIAL AND VIRAL BIOINFORMATICS RESOURCE CENTER

The Bacterial and Viral Bioinformatics Resource Center (BV-BRC) is an information system designed to support research on bacterial and viral infectious diseases via integration of vital pathogen information with rich data and analysis tools. BV-BRC combines two long-running centers: PATRIC, the bacterial system, and IRD/ViPR, the viral systems.

BROWSE ARCHAEA BACTERIA PHAGES VIRUSES EUKARYOTIC HOSTS

SEARCH All Data Types Find a gene, genome, microarray, etc. Q All terms

PATRIC

The Pathosystems Resource Integration Center (PATRIC) provides an extensive collection of integrated bacterial data, tools, and visualizations for genomic analysis to aid researchers in discovery and characterization of mechanisms of pathogenicity, supporting development of diagnostics, therapeutics, and vaccines.

[Quickstart Video](#)
[User Guides](#)
[Tutorials](#)
[Common Tasks](#)
[CLI Tutorial](#)
[Instructional Videos](#)

IRD AND VIPR

The Influenza Research Database (IRD) and Virus Pathogen Resource (ViPR) provide a wide range of data, analysis and visualization tools, and personal workbenches for the virus research communities that facilitate understanding of viral pathogens and how they interact with their host, leading to new diagnostics, treatments and preventions.

[IRD Help Manual](#)
[IRD Tutorials & Training Methods](#)
[Contact IRD](#)
[ViPR Help Manual](#)
[ViPR Tutorials & Training Materials](#)
[Contact ViPR](#)

NEWS & ANNOUNCEMENTS

[Bioinformatics Resource Centers \(BRCs\) Impact Survey](#)

This survey will help the National Institute of Allergy and Infectious Diseases (NIAID) understand the usage and utility of the ... [read more](#)

Bioinformatics Resource Centers (BRCs) Impact Survey

BV-BRC
A Twitter list by @BVBRC_DB

VIPR BRC
@vipr_db

Only modest difference in vaccine effectiveness were noted with the delta variant as compared with the alpha variant after a shot of two vaccine doses. [nejm.org](#) 10...

Uploads Jobs 0 0 3119

2. A complete list of all completed jobs will appear from most recent to the very first job ever submitted. Clicking on any of the column heads will resort the page to show the results in that order.

BV-BRC BETA ORGANISMS DATA SEARCHES SERVICES WORKSPACES All Data Types Find a gene, genome, microarray, etc

Job Status
Last updated: 1:13:42

All Services 0 queued 0 running 3119 completed 199 failed

Status	ID	Service	Output Name	Submit	Start	Completed
completed	3925566	TaxonomicClassification	SRR1778456 human gut metagenome	7/23/21, 10:22 AM	7/23/21, 10:36 AM	7/23/21, 10:47 AM
completed	3925454	TaxonomicClassification	ERR3201760 sewage metagenome Tax Class	7/23/21, 8:24 AM	7/23/21, 10:29 AM	7/23/21, 10:30 AM
completed	3925452	MetagenomeBinning	ERR3201760	7/23/21, 8:23 AM	7/23/21, 10:28 AM	7/23/21, 12:40 PM
completed	3925440	MetagenomeBinning	SRR14999246 lung transplant	7/23/21, 7:26 AM	7/23/21, 10:25 AM	7/23/21, 10:26 AM
completed	3924850	CodonTree	Steno_tree_100	7/22/21, 4:54 PM	7/22/21, 6:44 PM	7/22/21, 6:47 PM
completed	3924703	CodonTree	Steno test_10 genes	7/22/21, 3:23 PM	7/22/21, 3:41 PM	7/22/21, 4:18 PM
completed	3924600	GenomeAssembly2	SRR14999246_assembl	7/22/21, 10:41 AM	7/23/21, 6:46 AM	7/23/21, 7:17 AM
completed	3924594	MetagenomeBinning	SRR14999246	7/22/21, 9:51 AM	7/23/21, 7:48 AM	7/23/21, 7:49 AM
completed	3922307	Proteome Comparison	Escherichia phage test	7/14/21, 2:08 PM	7/14/21, 2:08 PM	7/14/21, 2:09 PM

HIDE

3. Clicking on an individual job or row will show possible downstream functions, which appear as icons, in the vertical green bar to the right of the list. Clicking on the View icon will rewrite the page to show the results of the selected job.

Job Status
Last updated: 3:18:53

All Services 0 queued 0 running 3205 completed 203 failed

Status	ID	Service	Output Name	Submit	Start	Completed
completed	3929667	GenomeAssembly2	Ni_aligned_Unicycler_2	7/29/21, 10:45 AM	7/30/21, 10:34 AM	7/30/21, 10:35 AM
completed	3929666	GenomeAssembly2	Ni_aligned_reads_SPAd	7/29/21, 10:44 AM	7/30/21, 10:33 AM	7/30/21, 10:34 AM
completed	3929655	FastqUtils	Ni_align_29July2021	7/29/21, 10:27 AM	7/29/21, 10:28 AM	7/29/21, 10:31 AM
completed	3928799	Annotation	Brucella test for tree	7/28/21, 4:19 PM	7/28/21, 4:20 PM	7/28/21, 4:29 PM
completed	3928614	GenomeAssembly2	Ni_Metaspades_27July2	7/28/21, 10:42 AM	7/28/21, 12:08 PM	7/28/21, 12:10 PM
completed	3928612	GenomeAssembly2	S4_Metaspades_27July	7/28/21, 10:39 AM	7/28/21, 12:02 PM	7/28/21, 1:40 PM
completed	3928610	GenomeAssembly2	S3_Metaspades_27July	7/28/21, 10:39 AM	7/28/21, 10:41 AM	7/28/21, 12:17 PM

HIDE
VIEW
REPORT ISSUE...

4. The files produced by the specific job will be shown on the Jobs results page. As with the Jobs page, clicking on an individual row will populate the vertical green bar with possible action icons, like viewing or downloading the data.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3926799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:26 PM
Run time	8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.contigs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_protein.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.txt	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.tar.gz	1.9 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.txt	4.6 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.xls	5.1 MB	me	Only me	7/28/21, 4:26 PM
GenomeReport.html	112.1 kB	me	Only me	7/28/21, 4:26 PM
genome_quality_details.txt	28.0 kB	me	Only me	7/28/21, 4:26 PM
load_files		me	Only me	7/28/21, 4:26 PM
quality.json	310.0 kB	me	Only me	7/28/21, 4:26 PM

5. Completed jobs can also be access through the workspace, which you can access by clicking on the **Workspaces** tab, which is at the top of any BV-BRC page.

BV-BRC BETA ORGANISMS DATA SEARCHES SERVICES WORKSPACES HELP

NEW: Check out our SARS-CoV-2 Variants of Concern and Genome Assembly and Annotation Service.

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BROWSE ARCHAEA BACTERIA PHAGES VIRUSES EUKARYOTIC HOSTS

SEARCH All Data Types Find a gene, genome, microarray, etc. Q All terms

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Quickstart Video
User Guides
Tutorials
Common Tasks
CLI Tutorial
Instructional Videos

IRD AND ViPR

The Influenza Research Database (IRD) and Virus Pathogen Resource (ViPR) provide a wide range of data, analysis and visualization tools, and personal workbenches for the virus research communities that facilitate understanding of viral pathogens and how they interact with their host, leading to new diagnostics, treatments and preventions.

IRD Help Manual
IRD Tutorials & Training Methods
Contact IRD
ViPR Help Manual
ViPR Tutorials & Training Materials
Contact ViPR

NEWS & ANNOUNCEMENTS

[Bioinformatics Resource Centers \(BRCs\) Impact Survey](#)

This survey will help the National Institute of Allergy and Infectious Diseases (NIAID) understand the usage and utility of the ... [read more](#)

Bioinformatics Resource Centers (BRCs)

Impact Survey

BVBRC

A Twitter list by @BVBRC_DB

ViPR BRC @vpr_db

"Only modest differences in vaccine effectiveness were noted with the delta variant as compared with the alpha variant after the receipt of two vaccine doses." [nejm.org/doi/full/10.10...](#)

Uploads Jobs 0 0 3119

6. This will open a drop-down box for the workspace. To view the home workspace, click on **home**.

BV-BRC BETA ORGANISMS DATA SEARCHES SERVICES WORKSPACES HELP

NEW: Check out our [SARS-CoV-2 Variant](#)

BACTERIAL AND VIRAL BIOINFORMATICS RESOURCE CENTER

The Bacterial and Viral Bioinformatics Resource Center (BV-BRC) is an information system...

Workspaces
home
Genome Groups
Feature Groups
Experiment Groups
Public Workspaces

My Jobs
My Genomes

7. This will rewrite the page to show the home directory. Scrolling down the page will show the files and folders in the workspace.

BV-BRC BETA ORGANISMS DATA SEARCHES SERVICES WORKSPACES All Data Types Find a gene, genome, microarray, et

ARWattam / home (642 items)

Name	Size	Owner	Members	Created
Parent folder	-	-	-	-
MRSP8468v4.fasta	2.6 MB	me	Only me	11/15/19, 11:40 AM
01-S1.fasta	5.4 MB	me	Only me	10/26/20, 1:46 PM
01_Yellow_0.fa	97.3 MB	me	Only me	11/4/20, 4:26 PM
02_Yellow_1.fa	50.8 MB	me	Only me	11/4/20, 4:34 PM
03_Green_0.fa	31.7 MB	me	Only me	11/4/20, 4:35 PM
04_June2020 Demo		me	Only me	6/4/20, 3:15 PM
04_Green_1.fa	49.0 MB	me	Only me	11/4/20, 4:36 PM
05_Blue_0.fa	46.6 MB	me	Only me	11/4/20, 4:37 PM
06_Blue_1.fa	174.8 MB	me	Only me	11/4/20, 4:38 PM

8. If the job is in a particular folder, that can be opened by double clicking on the row that has the job.

ARWattam / home (646 items)

Name	Size	Owner	Members	Created
Taxonomic B...		me	Only me	4/4/19, 10:29 AM
Taxonomic C...ion test		me	Only me	3/26/19, 11:07 AM
Test folder		me	Only me	10/22/17, 9:44 AM
Test_01_Yellow_0.fa	52.5 MB	me	Only me	11/5/20, 9:59 AM
Test_Rockhopper_208964.12_gene_exp.gmx	363.9 kB	me	Only me	1/26/18, 11:32 AM
Thailand		me	Only me	5/15/18, 7:45 AM
Tn_Seq		me	Only me	11/7/17, 10:24 AM
Trash Can		me	Only me	11/2/17, 2:37 PM
Treponema		me	Only me	5/24/19, 6:45 AM

9. This rewrites the page to show the data in that particular folder. Completed jobs are indicated by a checked flag in the first column. The completed job can be accessed by double clicking on that row.

ARWattam / home / Test folder (242 items)

Name	Size	Owner	Members	Created
Brucella test for tree	32.0 kB	me	Only me	7/28/21, 4:28 PM
Brucella test mauve	4.8 kB	me	Only me	7/16/19, 3:08 PM
Brucella test of beta	4.6 kB	me	Only me	12/18/18, 7:43 AM
Buchnera aphidicola SRR4240359 test ATCC	7.3 kB	me	Only me	7/25/18, 12:45 PM
Buchnera G002 and Bag	3.4 kB	me	Only me	10/22/18, 4:49 PM
Buchnera aphidicola SRR4240359_CGA_Webinar	32.9 kB	me	Only me	8/8/18, 5:43 PM
Buchnera aphidicola SRR4240359_fastq	32.2 kB	me	Only me	5/3/18, 12:44 PM
Buchnera aphidicola SRR4240359_number	32.3 kB	me	Only me	5/3/18, 12:51 PM
Buchnera aphidicola Tuc7	30.7 kB	me	Only me	5/1/18, 1:40 PM
Buchnera aphidicola str. Tuc7	7.1 kB	me	Only me	4/25/18, 12:20 PM
Buchnera aphidicola_Aphis gossypii_PASI strain	7.6 kB	me	Only me	2/26/21, 12:31 PM

10. This will rewrite the page to show the files produced by the completed job. Clicking on an individual row will populate the vertical green bar with possible action icons, like viewing or downloading the data. Every annotation job (bacterial, bacteriophage and viral) includes **contigs.fasta**, **embl**, **feature_dna.fasta**,

feature_protein.fasta, **features.txt**, **gb**, **genome**, **gff**, **merged.gb**, **tar.gz**, **txt**, **xls** and **quality.json** files, as well as a **load_files** folder.

[Link to Genome Landing Page](#)

ARWattam / home / Test folder / Murine coronavirus test AM2 (24 Items)

GenomeAnnotation Job Result

Genome	Feature count (27), Organism (Murine coronavirus test AM2), Domain (Viruses), Annotation ID (694005.1454)				
Job ID	17824489				
Start time	12/3/24, 5:14 PM				
End time	12/3/24, 5:54 PM				
Run time	40m10s				
Parameters					

Name	Size	Owner	Members	Created
Parent folder	-	-	-	-
Murine coronavirus test AM2.contigs.fasta	31.9 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.embl	81.0 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.feature_dna.fasta	69.4 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.feature_protein.fasta	24.7 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.features.txt	3.3 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.gb	78.5 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.genome	93.2 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.gff	2.8 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.merged.gb	78.5 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.tar.gz	20.5 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.txt	91.0 kB	me	Only me	12/3/24, 5:54 PM
Murine coronavirus test AM2.xls	98.8 kB	me	Only me	12/3/24, 5:54 PM
load_files	-	me	Only me	12/3/24, 5:54 PM
quality.json	740 B	me	Only me	12/3/24, 5:54 PM

Annotations:

- Murine coronavirus test AM2.contigs.fasta** = assembled contigs in DNA FASTA format
- Murine coronavirus test AM2.embl** = EMBL dump of the annotated genome
- Murine coronavirus test AM2.feature_dna.fasta** = feature sequences of the genome in DNA FASTA format
- Murine coronavirus test AM2.feature_protein.fasta** = feature sequences of the genome in protein FASTA format
- Murine coronavirus test AM2.features.txt** = tab-delimited text file listing all the features of the genome
- Murine coronavirus test AM2.gb** = annotated genome in GenBank format
- Murine coronavirus test AM2.genome** = json format file that encapsulates all genome data
- Murine coronavirus test AM2.gff** = all genome features in General Feature Format
- Murine coronavirus test AM2.merged.gb** = GenBank file with individual contigs in GenBank fasta record
- Murine coronavirus test AM2.tar.gz** = zipped file containing all the annotation job information
- Murine coronavirus test AM2.txt** = text file with nucleotide and protein sequence of all genes
- Murine coronavirus test AM2.xls** = excel file with nucleotide and protein sequence of all genes
- load_files** = contains all the json files related to the annotation job
- quality.json** = contains all the json files related to the job in json format

Annotation job results

- Any annotation job run in the BV-BRC contains a number of files, as well as information about the submitted job. To view the input parameters that were selected when the job was submitted, click on the arrow that precedes the word **Parameters**.

ARWattam / home / Test folder / Brucella test for tree (16 Items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)				
Job ID	3928799				
Start time	7/28/21, 4:20 PM				
End time	7/28/21, 4:28 PM				
Run time	8m1s				
Parameters					

Name	Size	Owner	Members	Created
Parent folder	-	-	-	-
Brucella test for tree.contigs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM

- This will open a drop-down box that shows the parameters. This box can be closed by clicking on the same arrow.

Parameters

```
{
  "skip_indexing": "0",
  "queue_nowait": "0",
  "taxonomy_id": "234",
  "scientific_name": "Brucella test for tree",
  "contigs": "/ARWattam@patricbrc.org/home/Brucella MOOC_Test_Rog-Annotate.contigs.fasta",
  "domain": "Bacteria",
  "output_file": "Brucella test for tree",
  "output_path": "/ARWattam@patricbrc.org/home/Test folder",
  "skip_workspace_output": "0",
  "public": "0",
  "reference_genome_id": "",
  "recipe": "default",
  "code": "11"
}
```

3. The **contigs.fasta** contains the assembled contigs of the genome in DNA FASTA format.

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3928799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder	-	-	-	-
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_protein.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM

4. The file ending in **.embl** contains an EMBL dump of the annotated genome.

[illegible]

5. The **feature_dna.fasta** file contains all the feature sequences of the genome in DNA FASTA format.

[illegible]

6. The **feature_protein.fasta** contains all the protein sequences of the genome in protein FASTA format.

[illegible]

7. The **features.txt** is a tab-delimited text file listing all the features of the genome. For each feature, it contains the BV-BRC ID, the location string, the feature type, the functional assignment, any alternated IDs found, and (for protein-coding genes) the protein MD5 [29] checksum.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM

8. The **gb** file contains the annotated genome in GenBank format.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM

9. The **genome** file contains a special "Genome Typed Object (GTO)" JSON-format file that encapsulates all the data from the annotated genome.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM

10. The **gff** lists all the features of the genome in General Feature Format.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

End time: 7/28/21, 4:28 PM

Run time: 8m1s

Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome: Feature count (3736), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)

Job ID: 3928799

Start time: 7/28/21, 4:20 PM

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Parameters

Name	Size	Owner	Members	Created
Parent folder				
Brucella test for tree.configs.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
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Brucella test for tree.feature_prot.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.tbl	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM

11. The **merged.gb** is a GenBank file where the individual contigs are each included as a GenBank fasta record under a single locus.

12. The **tar.gz** file is a zipped file that contains all the information about the annotation job.

13. The text, or **txt** file shows the nucleotide and protein sequence of all the annotated genes.

14. The **.xls** is an excel file that shows the nucleotide and protein sequence of all the annotated genes.

15. The **genome_quality_details.txt** file is only produced when the **Bacteria/Archaea** recipe was selected. It shows some of the quality scores seen in the **GenomeReport.html**, and a list of the genes in both the newly annotated and reference genomes, and the number of copies of this gene in each.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3628799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s

Parameters

Name	Size	Owner	Members	Created
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_protein.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.txt	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.tar.gz	1.9 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.bd	4.6 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.xls	5.1 MB	me	Only me	7/28/21, 4:26 PM
GenomeReport.html	112.1 kB	me	Only me	7/28/21, 4:26 PM
genome_quality_details.txt	28.0 kB	me	Only me	7/28/21, 4:26 PM
load_files		me	Only me	7/28/21, 4:26 PM

txt file: genome_quality_details.txt

```

Good Seed: Y
Completeness: 100
Contamination: 3.1
Group: R200 (Brucella melitensis bv. 1 str. 16M)
AbcTranInvoCytoC2 1 1
AcetCoenCarbTran2 1 2
AspaTrnaAmidSubu 1 1
AtpSyntF0SectSubu2 1 1
CellDiviProtDivi 1 1
CytoCHEMElyasSubu2 1 1
CytoCOxidSubuCoq2 1 1
SnaBindProtHusBeta 1 1
EndoIii 1 1
Fig0HypoProt1922 1 1
Fig0HypoProt3577 1 1
Fig0HypoWithDnaJ 1 1
Fig1GluDepeThio 1 1
GluTrnaAmidSubu 1 1
GlycCleasSystHProt 1 1
GtpBindProtEra 1 1
HypoProtDuf2a3 1 1
HypoProtThatOfte 1 1
ImidPhosBehy 1 1

```

16. The **Load** /files folder contains all the json files related to the annotation job.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3628799
Start time	7/28/21, 4:20 PM
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Parameters

Name	Size	Owner	Members	Created
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Brucella test for tree.feature_protein.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.txt	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.tar.gz	1.9 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.bd	4.6 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.xls	5.1 MB	me	Only me	7/28/21, 4:26 PM
GenomeReport.html	112.1 kB	me	Only me	7/28/21, 4:26 PM
genome_quality_details.txt	28.0 kB	me	Only me	7/28/21, 4:26 PM
load_files		me	Only me	7/28/21, 4:26 PM
quality.json	310.0 kB	me	Only me	7/28/21, 4:26 PM

ARWattam / home / Test folder / Brucella test for tree / load_files (9 items)

Name	Size	Owner	Members	Created
Parent folder	-	-	-	-
feature_sequence.json	4.8 MB	me	Only me	7/28/21, 4:26 PM
genome.json	1.4 kB	me	Only me	7/28/21, 4:26 PM
genome_amr.json	3 B	me	Only me	7/28/21, 4:26 PM
genome_feature.json	3.5 MB	me	Only me	7/28/21, 4:26 PM
genome_sequence.json	3.3 MB	me	Only me	7/28/21, 4:26 PM
pathway.json	1.6 MB	me	Only me	7/28/21, 4:26 PM
sp_gene.json	496.4 kB	me	Only me	7/28/21, 4:26 PM
subsystem.json	1.7 MB	me	Only me	7/28/21, 4:26 PM
taxonomy.json	3 B	me	Only me	7/28/21, 4:26 PM

17. The **quality.json** file has the same information in json format.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3628799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s

Parameters

Name	Size	Owner	Members	Created
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_protein.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.txt	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.tar.gz	1.9 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.bd	4.6 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.xls	5.1 MB	me	Only me	7/28/21, 4:26 PM
GenomeReport.html	112.1 kB	me	Only me	7/28/21, 4:26 PM
genome_quality_details.txt	28.0 kB	me	Only me	7/28/21, 4:26 PM
load_files		me	Only me	7/28/21, 4:26 PM
quality.json	310.0 kB	me	Only me	7/28/21, 4:26 PM

json file: quality.json

```

{
  "amr_gene_summary": {
    "antibiotic inactivation enzyme": [
      "FosX"
    ],
    "antibiotic target in susceptible species": [
      "Alr",
      "Ddl",
      "EF-G",
      "EF-Tu",
      "folA, Dfr",
      "folP",
      "gyrA",
      "gyrB",
      "inhA, fabI",
      "Iso-tRNA",
      "kdsA",
      "MurA",
      "rho"
    ]
  }
}

```

18. Genomes annotated by VIGOR4, will include additional files in the results folder, as seen below.

	load_files
	quality.json
	vigor4.stderr.txt
	vigor4.stdout.txt
	vigor_out-20241203-193205.ini
	vigor_out.aln = Alignment of predicted protein to reference, and reference protein to genome
	vigor_out.cds = Fasta file of predicted CDSs
	vigor_out.gff3 = Predicted features in GFF3 format
	vigor_out.pep = Fasta file of predicted proteins
	vigor_out.rpt = Summary of program results
	vigor_out.tbl = Predicted features in GenBank tbl format
	vigor_out.warnings

Genome Report

Genome quality analysis is automatically performed when using the Genome Annotation (Metagenomic Binning Service) for bacterial and archaeal genomes. The genome quality tools look at the functional roles (or genes) present in an annotated genome to determine if the genome looks correct. Two separate mechanisms are used to predict the number of times each gene should be found in the genome. A role is good if it occurs the predicted number of times; otherwise it is problematic.

The first quality tool (EvalG) checks the completeness and contamination of the genome using a re-implementation of the CheckM [30] algorithm. EvalG identifies universal genes that are expected to occur exactly once in all genomes of a particular taxonomic grouping. Missing genes indicate the genome is less complete; extra genes indicate the genome may be contaminated.

The second quality tool (EvalCon) [31] checks the consistency of the genome annotation. Over 1300 genes that have a predictable relationship to other genes were identified by a machine learning process. EvalCon determines how many instances of each gene are expected given the list of other genes in the genome. If a gene in the genome is unexpected, or an expected gene is missing, this is considered coarse inconsistency. If a gene occurs a different number of times than predicted, this is fine inconsistency.

The four numbers –completeness, contamination, coarse consistency, and fine consistency– express measures of the quality of the genome. EvalG and EvalCon relay not only the numbers, but also identify the problematic genes that occur an unexpected number of times. These are summarized in the Quality Report web page. The report is divided into three sections– the Summary Section that describes the genome itself, the Problematic Roles Report that lists the genes whose expected and actual occurrence numbers do not match, and the Contig Report that lists the contigs containing problematic roles.

1. To view the **GenomeReport.html**, click on its row and then on the View icon.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3928799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s
Parameters	

Name	Size	Owner	Members	Created
Brucella test for tree.embl	7.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_dna.fasta	3.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.feature_protein.fasta	1.3 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.features.txt	491.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.genome	12.4 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.gff	521.9 kB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.merged.gb	7.2 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.tar.gz	1.9 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.txt	4.6 MB	me	Only me	7/28/21, 4:26 PM
Brucella test for tree.xls	5.1 MB	me	Only me	7/28/21, 4:26 PM
GenomeReport.html	112.1 kB	me	Only me	7/28/21, 4:26 PM
genome_quality_details.txt	28.0 kB	me	Only me	7/28/21, 4:26 PM
load_files		me	Only me	7/28/21, 4:26 PM

VIEW CDS BROWSER

HIDE

GUIDE

DOWNLOAD

VIEW

DELETE

RENAME

COPY

MOVE

EDIT TYPE

2. This will reload the page to show the **Genome Report**. The top of the page is the summary section, which shows the scores achieved by the binned genome.

html file: GenomeReport.html 

Results for 234.551

[Tutorial for this page.](#)

Genome ID	234.551
Genome Name	Brucella test for tree
Reference Genomes	29459.406
Coarse consistency (%)	99.9
Fine consistency (%)	94.0
Completeness (%)	100
Contamination (%)	3.1
Evaluation Group	R200 (Brucella melitensis bv. 1 str. 16M)
Contig count	2
DNA size (bp)	3325229
Contigs N50 (bp)	2119916
Contigs L50	1
Overpresent Roles	78
Underpresent Roles	4
Predicted Roles	1369
Completeness Roles	31
Total Distinct Roles	2623
Protein-Encoding Genes with Functional Assignment	2648
Protein-Encoding Genes without Functional Assignment	932
% Protein-Encoding Feature Coverage	107.66
% Features that are Hypothetical	26.03
% Features that are in Local Protein Families	96.45

3. A yellow color indicates a poor scoring value.

Genome ID	1247024.51
Genome Name	Arsenophonus endosymbiont clonal population
Reference Genomes	
Coarse consistency (%)	94.4
Fine consistency (%)	93.8
Completeness (%)	100
Contamination (%)	7.7
Evaluation Group	root
Contig count	1419
DNA size (bp)	4757144
Contigs N50 (bp)	5640
Contigs L50	158
Overpresent Roles	46
Underpresent Roles	38
Predicted Roles	1357
Completeness Roles	24
Total Distinct Roles	845
Protein-Encoding Genes with Functional Assignment	792
Protein-Encoding Genes without Functional Assignment	8944
% Protein-Encoding Feature Coverage	204.66
% Features that are Hypothetical	91.87
% Features that are in Local Protein Families	6.45

Potentially problematic roles in 1720349.53

Potentially problematic roles in 1720349.53

5/13/25, 11:54 AM

has four columns, including the **Contig Name**, the **Length** of the contig, the number of **Good Features** (or genes) on that contig, and link to a table with those problematic genes (**Feature link**).

Potentially Problematic Roles by Contig

Contig Name	Length (bp)	Good Features	Feature link
assembly_contig_2	696068	104	12 features
assembly_contig_3	679581	194	7 features
assembly_contig_4	609544	175	6 features
assembly_contig_5	545709	113	5 features
assembly_contig_8	480886	160	11 features
assembly_contig_9	425805	133	3 features
assembly_contig_13	349588	81	1 feature
assembly_contig_15	338167	149	5 features
assembly_contig_16	316143	118	1 feature
assembly_contig_17	308540	88	4 features
assembly_contig_20	256361	16	2 features
assembly_contig_22	228539	30	4 features
assembly_contig_23	222497	98	3 features
assembly_contig_25	194100	66	3 features

Viewing the Annotated Genome

Private genomes that have been annotated in BV-BRC can be viewed directly from the annotation job, or through the workspace, or by using the Global Search function.

Viewing the genome from the job report

1. At the top right of the annotation job there are three icons that provide a direct link to information about the genome. Clicking on the **View** icon will open a new tab that contains the Genome landing page, with all the information about the newly annotated genome.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3928799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s

VIEW CDS BROWSER

Genome View

Bacteria » Proteobacteria » Alphaproteobacteria » Rhizobiales » Brucellaceae » Brucella » Brucella test for tree

Overview AMR Phenotypes Phylogeny Genome Browser Circular Viewer Sequences Features Specialty Genes Protein Families Pathways Subsystem

Brucella test for tree Edit Genomic Features

Length: 3325229bp, Contigs: 2

Organism Info

Genome ID	234.551
Genome Name	Brucella test for tree
NCBI Taxon ID	234
Genome Status	WGS

Genome Quality

Genome Quality	Good
Coarse Consistency	99.9
Fine Consistency	94

Sharing

None available

Isolate Info

None available

Host Info

None available

Sequence Info

Contigs	2
Genome Length	3325229
GC Content	57.24027

Genomic Features

	PATRIC	RefSeq
CDS	3580	0
repeat_region	94	0
tRNA	55	0
rRNA	9	0

Protein Features

	PATRIC	RefSeq
Hypothetical proteins	653	0
Proteins with functional assignments	2927	0
Proteins with EC number assignments	0	0
Proteins with GO assignments	869	0
Proteins with Pathway assignments	0	0
Proteins with Subsystem assignments	0	0
Proteins with PATRIC genus-specific family (PLfam) assignments	3453	0
Proteins with PATRIC cross-genus family (PGfam) assignments	3478	0

2. Clicking on the **CDS** icon will open a new tab that contains a table with all of the genes that have been annotated in the genome.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (4.551)
Job ID	3928799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s

Red arrow pointing to the 'CDS' icon in the top right corner of the ARWattam interface.

Genome View

Bacteria » Proteobacteria » Alphaproteobacteria » Rhizobiales » Brucellaceae » Brucella » Brucella test for tree

Overview AMR Phenotypes Phylogeny Genome Browser Circular Viewer Sequences **Features** Specialty Genes Protein Families Pathways Subsy

DOWNLOAD KEYWORDS CDS PATRIC FEATURE_TYPE x ANNOTATION x FILTERS

☐	PATRIC ID	RefSeq Locus Tag	Gene Symbol	PATRIC Local Family	PATRIC Global Family	Product
☐	fgl234.551.peg.1					hypothetical protein
☐	fgl234.551.peg.2			PLF_234_00002217	PGF_01030732	FIG00450939: hypothetical protein
☐	fgl234.551.peg.3			PLF_234_00001776	PGF_04729382	hypothetical protein
☐	fgl234.551.peg.4			PLF_234_00001597	PGF_06463602	Glutathione S-transferase (EC 2.5.1.18)
☐	fgl234.551.peg.5			PLF_234_00002936	PGF_01030679	FIG00450180: hypothetical protein
☐	fgl234.551.peg.6			PLF_234_00003042	PGF_01030945	FIG00450262: hypothetical protein
☐	fgl234.551.peg.7			PLF_234_00000765	PGF_00013801	AMP nucleosidase (EC 3.2.2.4)
☐	fgl234.551.peg.8			PLF_234_00000765	PGF_00013801	AMP nucleosidase (EC 3.2.2.4)
☐	fgl234.551.peg.9			PLF_234_00002243	PGF_00014057	AP endonuclease family 1 domain protein
☐	fgl234.551.peg.10			PLF_234_00000709	PGF_00424123	Electron transfer flavoprotein-ubiquinone oxidoreductase (EC 1.5.5
☐	fgl234.551.peg.11			PLF_234_00001758	PGF_09218369	Uracil-DNA glycosylase, family 4 (EC 3.2.2.27)
☐	fgl234.551.peg.12			PLF_234_00000567	PGF_09996723	Uncharacterized MFS-type transporter
☐	fgl234.551.peg.13			PLF_234_00001000	PGF_00720080	Membrane protein
☐	fgl234.551.peg.14			PLF_234_00000890	PGF_00019484	Membrane alanine aminopeptidase N (EC 3.4.11.2)
☐	fgl234.551.peg.15			PLF_234_00000890	PGF_00019484	Membrane alanine aminopeptidase N (EC 3.4.11.2)
☐	fgl234.551.peg.16			PLF_234_00000890	PGF_00019484	Membrane alanine aminopeptidase N (EC 3.4.11.2)
☐	fgl234.551.peg.17			PLF_234_00001047	PGF_00050512	Two-component system sensor histidine kinase
☐	fgl234.551.peg.18			PLF_234_00000803	PGF_06736063	hypothetical protein
☐	fgl234.551.peg.19			PLF_234_00000654	PGF_01053024	Glutamine synthetase adenylyl-L-tyrosine phosphorylase (EC 2.7.7

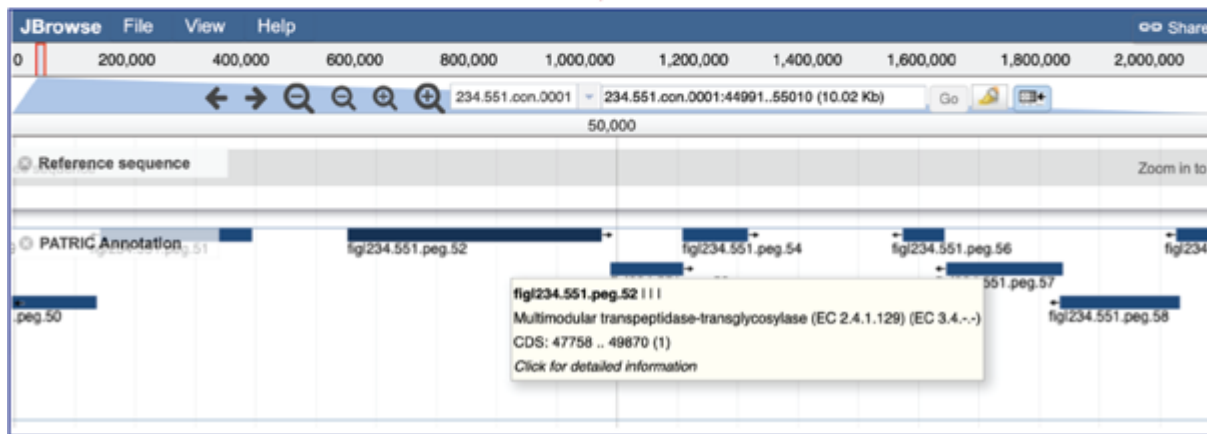
1 - 200 of 3580 results

3. Clicking on the **Browser** icon will open a new tab that contains the genome browser, showing all the genes and their orientation in the genome.

ARWattam / home / Test folder / Brucella test for tree (16 items)

GenomeAnnotation Job Result

Genome	Feature count (3738), Organism (Brucella test for tree), Domain (Bacteria), Annotation ID (234.551)
Job ID	3928799
Start time	7/28/21, 4:20 PM
End time	7/28/21, 4:28 PM
Run time	8m1s



Finding the genome using the workspace

1. Private genomes can be located through the workspace. Click on the **Workspaces** tab at the top of any page. This will open a drop-down box. Click on **My Genomes** at the upper right of this box.



2. This will open a table that contains all of the genomes that you have annotated. The top of the table contains a text box that can be used to filter the data. Enter the name, or the unique genome ID into this box and then hit return on your keyboard.

Genome List View
GENOMES: PUBLIC IS false (1198 Genomes)

Overview **Genome** AMR Phenotypes Sequences Features Specialty Genes Protein Families Pathways Subsystems Transcriptomics Interaction

DOWNLOAD KEYWORDS FILTERS APPLY

☐	Genome Name	Genome ID	Genome Status	Contigs	PATRIC CDS	Isolation Country	Host Name	Collection Year	Completion Date
☐	Staphylococcus aureus strain S1_S12	1280.17370	WGS	42	2693				
☐	Staphylococcus aureus strain S80_S2	1280.17440	WGS	84	2786				
☐	Nitrospina sp. UBA6737 clonal population	1946981.7	WGS	577	2295				
☐	Helicobacter pylori HP34	210.5494	WGS	42	1655				
☐	Corynebacterium sp. HMSC078A10 clonal population	1739312.39	WGS	50	2601				
☐	Bacteria SRR1955831-2-3_Unicycler_4_2.8279	WGS	2	2729					
☐	Clostridium perfringens clonal population	451754.15	WGS	163	2895				
☐	Oceanobacillus oncorhynchi clonal population	545501.6	WGS	29	4872				
☐	Serratia marcescens clonal population	615.1338	WGS	1186	7475				
☐	Coxiella burnetii Cb175_Guyana-HG8	777.212	WGS	1	2675				
☐	Staphylococcus aureus strain S74_S2	1280.17433	WGS	1331	3982				
☐	Staphylococcus aureus BS-S37	1280.17163	WGS	51	2709				
☐	Staphylococcus aureus BS-S62-RB	1280.17187	WGS	40	2644				
☐	Staphylococcus aureus BS-S2-RB	1280.17201	WGS	56	2771				
☐	uncultured Prevotella clonal population	159272.309	WGS	3223	5712				
☐	Escherichia coli clonal population	562.49301	WGS	780	6032				

3. This will filter the genomes to show those that match the text entered. Clicking on the row that contains the correct genome will highlight the vertical green bar to the right with possible downstream functions. Clicking on the **Genome** icon will open a new tab that has the Genome landing page for that genome.

Genome List View
GENOMES: PUBLIC IS false (1198 Genomes)

Overview **Genomes** AMR Phenotypes Sequences Features Specialty Genes Protein Families Pathways Subsystems Transcriptomics Interactions

DOWNLOAD KEYWORDS FILTERS APPLY

Brucella test

☐	Genome Name	Genome ID	Genome Status	Contigs	PATRIC CDS	Isolation Country	Host Name	Collection Year	Completion Date
☐	Brucella abortus test new system S5_L	29460.33	WGS	26	3292				
☐	Brucella abortus test new system S5_L	234.490	WGS	30	3306				
☒	Brucella test for tree	234.551	WGS	2	3580				

HIDE GUIDE DWNLD COPY GENOME GROUP

Finding the genome using Global Search

1. The Global Search can be used to locate public and private data. The name of the genome can be entered into the box, which can be found on the home page, and also at the top right of any other page.

BV-BRC ALPHA ORGANISMS DATA SEARCHES SERVICES WORKSPACES HELP ABOUT

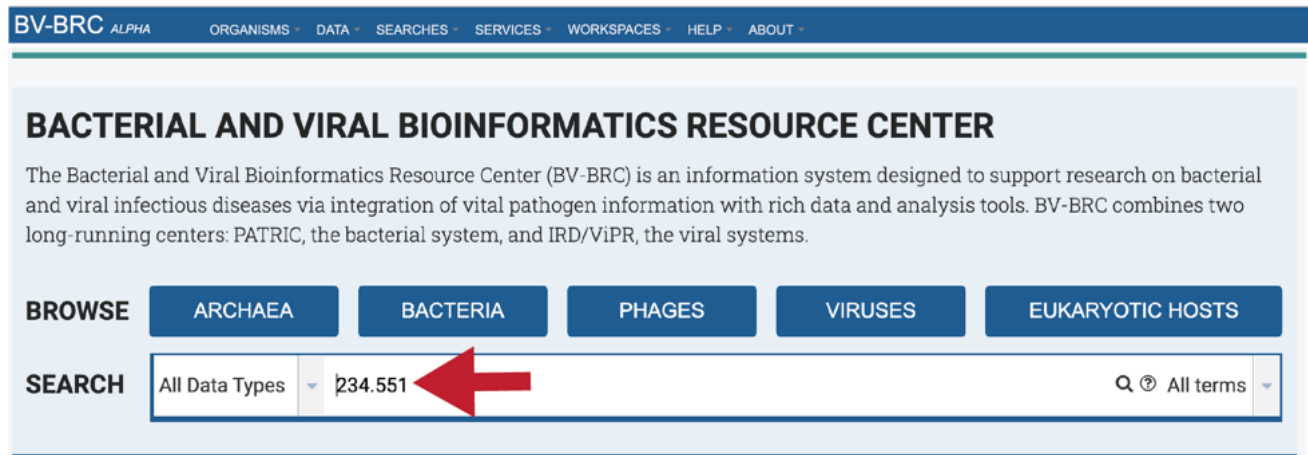
BACTERIAL AND VIRAL BIOINFORMATICS RESOURCE CENTER

The Bacterial and Viral Bioinformatics Resource Center (BV-BRC) is an information system designed to support research on bacterial and viral infectious diseases via integration of vital pathogen information with rich data and analysis tools. BV-BRC combines two long-running centers: PATRIC, the bacterial system, and IRD/ViPR, the viral systems.

BROWSE ARCHAEA BACTERIA PHAGES VIRUSES EUKARYOTIC HOSTS

SEARCH All Data Types All terms

2. Entering the Genome ID, if known, will provide a more direct access to an individual genome. Click return after entering.



BV-BRC ALPHA ORGANISMS DATA SEARCHES SERVICES WORKSPACES HELP ABOUT

BACTERIAL AND VIRAL BIOINFORMATICS RESOURCE CENTER

The Bacterial and Viral Bioinformatics Resource Center (BV-BRC) is an information system designed to support research on bacterial and viral infectious diseases via integration of vital pathogen information with rich data and analysis tools. BV-BRC combines two long-running centers: PATRIC, the bacterial system, and IRD/ViPR, the viral systems.

BROWSE ARCHAEA BACTERIA PHAGES VIRUSES EUKARYOTIC HOSTS

SEARCH All Data Types All terms

3. This will re-write the page to show the results of the search. Clicking on the name under Genomes will re-write the page to show the landing page for that genome.



"234.551"

Antibiotics: 0	Genomes: 1	Genomic Features: 3738
Specialty Genes: 0	Taxonomy: 0	Transcriptomics Experiments: 0

Top Matches

Genomes (1)

Brucella test for tree
Genome ID: 234.551 | 2 Contigs

Genomic Features (3738)

Integrase
Brucella test for tree
PATRIC | CDS | fig|234.551.peg.3164

hypothetical protein
Brucella test for tree
PATRIC | CDS | fig|234.551.peg.643

hypothetical protein
Brucella test for tree
PATRIC | CDS | fig|234.551.peg.897

References

1. Brettin, T. et al. RASTtk: A modular and extensible implementation of the RAST algorithm for building custom annotation pipelines and annotating batches of genomes. *Sci Rep* 5: 8365 (2015).
2. Wang, A. et al. VIGOR, an annotation program for small viral genomes. *BMC Bioinformatics* 11:451 (2010).
3. Wang, A. et al. VIGOR extended to annotate genomes for additional 12 different viruses. *Nucleic Acids Res* 40:W186-92 (2012).
4. Larsen, C.N. et al. Mat_peptide: comprehensive annotation of mature peptides from polyproteins in five virus families. *Bioinformatics* 36(5):1627-1628 (2020).
5. McNair, K. et al. Phage Genome Annotation Using the RAST Pipeline. *Methods Mol Biol*: 1681:231-238 (2018).
6. McNair, K. et al. PHANOTATE: a novel approach to gene identification in phage genomes. *Bioinformatics* 35, 4537-4542 (2019).

7. Aziz, R. K. et al. The RAST Server: rapid annotations using subsystems technology. *BMC genomics* 9, 75 (2008).
8. Chan, P.P. et al. tRNAscan-SE 2.0: improved detection and functional classification of transfer RNA genes. *Nucleic Acids Res* 49:9077–9096 (2021).
9. Altschul, S.F. et al. Basic local alignment search tool. *J. Mol. Biol.* 215:403-410 (1990).
10. Croucher, N.J. et al. Identification, variation and transcription of pneumococcal repeat sequences. *BMC genomics* 12, 1-13 (2011).
11. Hyatt, D. et al. Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC Bioinformatics*: 11:119 (2010).
12. Delcher, A. L. et al. Identifying bacterial genes and endosymbiont DNA with Glimmer. *Bioinformatics* 23: 673-679 (2007).
13. Schapire R.E. Explaining adaboost. In: *Empirical inference*. Springer; 2013. p. 37–52.
14. Kent, W. J. BLAT—the BLAST-like alignment tool. *Genome research* 12, 656-664 (2002).
15. Johnson, M. et al. NCBI BLAST: a better web interface. *Nucleic acids research* 36, W5-W9 (2008).
16. Liu, B. et al. VFDB 2022: a general classification scheme for bacterial virulence factors. *Nucleic Acids Res.* 50(D1):D912-D917 (2022).
17. He Y, et al. Updates on the web-based VIOLIN vaccine database and analysis system. *Nucleic Acids Res.* 42(Database issue):D1124-32 (2014).
18. Mao, C. et al. Curation, integration and visualization of bacterial virulence factors in PATRIC. *Bioinformatics* 31, 252-258 (2015).
19. Alcock, B.P. et al. CARD 2020: antibiotic resistome surveillance with the comprehensive antibiotic resistance database. *Nucleic Acids Res* 48(D1):D517-D525 (2020).
20. Davis, J.J. et al. Antimicrobial resistance prediction in PATRIC and RAST. *Scientific reports* 6, 27930 (2016).
21. Saier, M.H. et al. The Transporter Classification Database (TCDB): 2021 update. *Nucleic Acids Res* 49(D1):D461-D467 (2021).
22. Knox, C., et al. DrugBank 6.0: the DrugBank Knowledgebase for 2024. *Nucleic Acids Res.* 52(D1):D1265-D1275 (2024).
23. Zhou Y. et al. TTD: Therapeutic Target Database describing target druggability information. *Nucleic Acids Res* 52(D1):D1465-D1477 (2024).
24. Davis, J. J. et al. PATtyFams: Protein families for the microbial genomes in the PATRIC database. 7, 118 (2016).
25. Overbeek, R. et al. The subsystems approach to genome annotation and its use in the project to annotate 1000 genomes. *Nucleic acids research* 33, 5691-5702 (2005).
26. Overbeek, R. et al. The SEED and the Rapid Annotation of microbial genomes using Subsystems Technology (RAST). 42, D206-D214 (2013).
27. Jolley, K.A. et al. Open-access bacterial population genomics: BIGSdb software, the PubMLST.org website and their applications. *Wellcome Open Res* 3:124 (2018).
28. Akhter, S. et al. PhiSpy: a novel algorithm for finding prophages in bacterial genomes that combines similarity-and composition-based strategies. *Nucleic acids research* 40: e126e126 (2012).
29. Rivest, R. RFC1321: The MD5 message-digest algorithm (1992).
30. Parks, D. H. et al. CheckM: assessing the quality of microbial genomes recovered from isolates, single

cells, and metagenomes. *Genome research* 25, 1043-1055 (2015).

31. Parrello, B. et al. A machine learning-based service for estimating quality of genomes using PATRIC. *BMC Bioinformatics* 20(1):486 (2019).